

MOHAMMAD TANVIR RAHMAN, Ph.D.

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PROFESSIONAL SUMMARY

Lead-level software engineer with 12+ years of experience designing and delivering full-stack and backend systems across cloud and data-intensive environments. Strong background in engineering modernization, microservices, REST and gRPC APIs, CI/CD, cloud platforms, data modeling, and enterprise integration. Proven record partnering with cross-functional stakeholders, defining technical approaches, improving engineering practices, and delivering production systems in client-facing Agile environments.

TECHNICAL SKILLS

Leadership: Engineering standards, system design, architecture ownership, technical guidance, cross-team collaboration, mentoring

Languages: C#, Java, Python, JavaScript, TypeScript, SQL, T-SQL

Full Stack and Backend: .NET, .NET Core, ASP.NET MVC, REST APIs, Web API, gRPC, WCF, Spring Boot, Node.js, Angular, React, microservices

Cloud and DevOps: Azure, AWS, Docker, Kubernetes, Terraform, Azure DevOps, CI/CD

Data and Integration: PostgreSQL, SQL Server, DynamoDB, Snowflake, Kafka, Spark, ETL, NoSQL, enterprise system integration

Engineering Practices: Automated testing, documentation, troubleshooting, observability, Agile/Scrum, production support

PROFESSIONAL EXPERIENCE

Texas A&M AgriLife Extension Service | Software Developer IV

Houston, TX | Aug 2024 – Present

- Design and deliver end-to-end features for two production applications using C#, Node.js, Angular, REST APIs, Azure, ArcGIS, and PostgreSQL.
- Contribute to architecture, data modeling, and integration design for systems supporting real-time planning, risk, and sustainability analysis.
- Improve team productivity through adoption of AI-assisted engineering tools and exploration of new AI and software development approaches to enhance quality, productivity, and modularity.
- Strengthen engineering quality through CI/CD improvements in Azure DevOps, automated testing, and improved development practices.

Self-Employed | Freelance Full Stack Engineer

Houston, TX | Aug 2023 – Jul 2024

- Designed and built a reservation platform end to end, including service architecture, backend APIs, database design, booking workflows, invoice management, notifications, testing, and deployment.
- Owned architecture and implementation decisions with focus on maintainability, clear service boundaries, and reliable business workflows.

Capital One | Senior Full Stack Engineer (Contract)**Remote | Mar 2023 – Jun 2023**

- Engineered, tested, and supported software for a cloud governance platform that accelerated Snowflake adoption, cost management, and automation of operational processes.
- Delivered production services and platform features using Java, React, TypeScript, SQL, DynamoDB, Docker, Kubernetes, AWS services, gRPC, and Snowflake.

Nordstrom | Senior Software Engineer (Contract)**Seattle, WA | Jul 2021 – Dec 2022**

- Led modernization of FinTech applications from monoliths to microservices and migrated them from on-premises environments to cloud-based platforms for scalability and cost optimization.
- Built production backend services using Java, Spring Boot, C#, .NET, REST APIs, Docker, Kubernetes, Terraform, SQL Server, DynamoDB, and AWS.
- Automated CI/CD and deployment workflows with BCDR support, reducing environment setup time by 30%.
- Restructured data models and migrated data to DynamoDB, reducing operating expense by 50%.
- Performed code reviews, enforced development standards, and improved observability with dashboards and alerts using New Relic and Splunk.

Microsoft | Software Engineer (Contract)**Redmond, WA | Aug 2020 – Jun 2021**

- Developed scalable tools and backend services from the ground up using C#, Java, Python, Azure, Azure DevOps, Docker, and TypeScript.
- Supported high-volume streaming and analytics workflows using Spark, Kafka, HBase, Elasticsearch, and distributed cluster infrastructure.
- Built monitoring and automation tooling, improved observability, wrote tests for stability, and resolved production issues in live environments.

Cosasco | Software Development Engineer**Houston, TX | Mar 2018 – May 2020**

- Designed and developed enterprise software, backend services, and data ingestion platforms using C#, .NET Core, ASP.NET MVC, WPF, SQL Server, Entity Framework, XML, and REST APIs.
- Built integration solutions connecting internal platforms with customer operational systems using OPC, NiFi, Modbus, ETL pipelines, and service-based interfaces to ensure uninterrupted data flow.
- Developed cloud-based IIoT and monitoring solutions using AWS, Python, and service-based architectures.

EARLIER EXPERIENCE

- University of Houston, Texas, USA | Research / Teaching Assistant
- Abul Khair Group, Bangladesh | Deputy Manager (IT)
- Robi Axiata Ltd, Bangladesh | Specialist
- Vonair Software Services, Bangladesh | Software Engineer

EDUCATION

- Ph.D., Computer Science | University of Houston
- B.S., Computer Science & Engineering | Bangladesh University of Engineering & Technology